

SIMULATION-GROUNDED AGENTIC AI FOR MILITARY COURSE OF ACTION DEVELOPMENT

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ABSTRACT

Large Language Model (LLM) based planning tools are able to generate doctrinally fluent courses of action (COAs) rapidly, but operate on textual probabilities alone without models for the consequences of the proposed COAs. We present a decision support concept that grounds a multi-agent LLM planning pipeline, mapped to the Norwegian Army planning process, in a closed simulation loop: the agents task a constructive simulation natively through military simulation standards and refine their plans from the returned simulation logs, guided by a hybrid retrieval layer coupling a vector database with a knowledge graph. The loop's state-action trajectories also provide the data needed to train a world model, which we explore as a fast surrogate for screening candidate COAs before high-fidelity simulation.

1 INTRODUCTION AND BACKGROUND

Operational planning requires staff to balance doctrine, terrain, logistics, and adversary capabilities under intense time pressure to reach an optimal plan. Reinforcement learning can ground action selection through reward function but lacks human-interpretable alignment with doctrine (Kallfass et al. 2023). Conversely, generative frameworks such as COA-GPT 2.0, show that multi-agent Large Language Model (LLM) systems can assemble doctrinally coherent planning products in seconds (Goecks et al. 2025). However, an LLM has no model of the world it plans in (LeCun 2022) - it cannot account for fuel, terrain, timing, or attrition, nor distinguish an articulate plan from a viable one. COA-GPT 2.0 delegates wargaming to a simulator, but does not present details on how simulation results feed back into plan refinement and whether simulation interoperability standards are utilized.

We argue that the most effective and verifiable grounding for LLM planning agents today is a simulation loop. Our concept maps the processes of the Norwegian Army staff handbook (Hærens våpenskole 2021) and improves on prior generative planners through standards-based simulation coupling, an explicitly closed refinement loop, and hybrid retrieval. The loop also generates exactly the training data required to learn a world model serving as a fast inner evaluation loop for the COA Development Agents (Figure 1).

2 A SIMULATION-GROUNDED AGENT ARCHITECTURE

Multi-agent layers collaborate sequentially to build the operational picture: *terrain analysis agents* use vision-language models and environmental services to distill tactical maps into military geography products, which feed a *HybridRAG* knowledge layer coupling a vector database of doctrine and handbooks with a knowledge graph of the operational environment and the order of battle. Whereas retrieval in prior planners is purely vector-based (Goecks et al. 2025), the graph supports multi-hop relational queries (e.g., which units can mutually support one another) and filters doctrinally unsound maneuvers out of the agents' drafts before simulation assets are committed. Both have been implemented and tested by FFI.

Grounding of candidate plans happens in a closed simulation loop, making this loop explicit and standards-based. The agent system tasks the *SWAP* constructive simulation (Bruvoll et al. 2015), built on VR-Forces. The COAs follow the C2SIM standard (SISO 2020) and are interpreted by a multi-agent system which communicates with VR-Forces. Orders and entity-level reports flow in a machine-readable, system-independent format. The returned telemetry (routes actually taken, timing, attrition, logistics) drives the next refinement iteration and is archived in the HybridRAG layer, accumulating lessons across planning

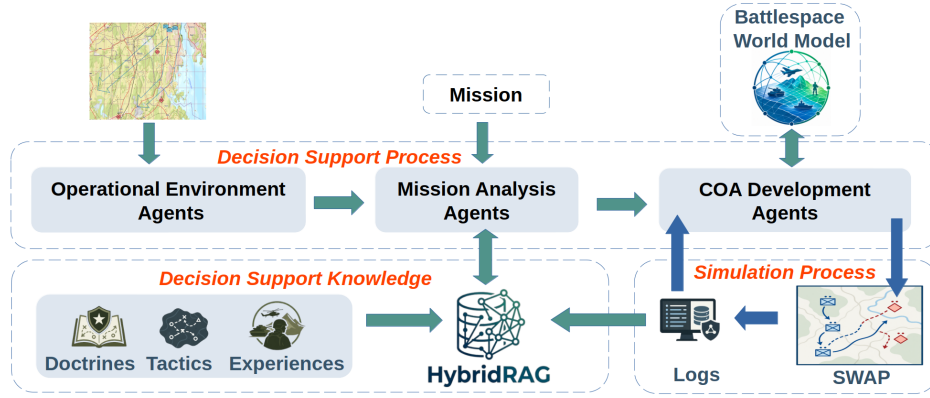


Figure 1: Concept sketch. Agents draft plans, ground them via SWAP/C2SIM telemetry, and optionally screen COAs with a learned world model.

sessions. The loop also accommodates the human planner, who can inject critiques or command constraints alongside the logs, prompting the agents to synthesize a more resilient plan.

3 OPPORTUNITY: LEARNING A WORLD MODEL FROM THE SIMULATION LOOP

High-fidelity simulation is the authoritative arbiter of physical feasibility, but too slow to call inside every reasoning step of an LLM agent. However, a by-product of the simulation is large volumes of state-action-outcome trajectories, exactly the data required to train a *world model* (LeCun, Yann 2022): a learned surrogate transition function $f : (s_t, a_t) \rightarrow \hat{s}_{t+1}$ mapping a numerical battlespace state s_t and a COA step a_t to a predicted next state. Operating on compressed latent states, the surrogate could roll out candidate maneuvers over extended horizons in a model-predictive control (MPC) fashion, discarding unsound options in milliseconds and reserving full simulation for feasible plans. World models have shown strong results in domains that, like combat simulation, mix continuous dynamics with discrete events (Hafner et al. 2025); to our knowledge, no COA development tool yet exploits a learned simulation surrogate inside its planning loop.

The world model is an early-stage direction, not a validated capability - open questions remain on state representation, fidelity bounds, and uncertainty communication - and appears as an optional inner loop in Figure 1. In summary, the agentic pipeline and HybridRAG layer are tested, the standards-based simulation loop is under integration, and the world model marks the research frontier that the first two make reachable.

REFERENCES

- Bruvoll, S., J. E. Hannay, G. K. Svendsen, M. Asprusten, K. M. Fauske, V. B. Kvernelv, *et al.* 2015. “Simulation-Supported Wargaming for Analysis of Plans”. In *M&S support to operational tasks including war gaming, logistics, cyber defence*, Volume 133 of *STO-MP-MSG*. Munich, Germany: NATO STO.
- Goecks, V. G., N. R. Waytowich, R. Bookman, and C. Kenny. 2025. “COA-GPT 2.0: An Agentic AI Planning Tool for Accelerating the Military Decision-Making Process”. Preprint. Under Review.
- Hærens våpenskole 2021. *Stabshåndbok for Hæren: Plan- og beslutningsprosessen*. Rena: Hæren. In Norwegian.
- Hafner, D., J. Pasukonis, J. Ba, and T. Lillicrap. 2025. “Mastering Diverse Control Tasks Through World Models”. *Nature* 640:647–653.
- Kallfass, D., T. Doll, and D. Kunde. 2023. “Doctrines into an AI Agent via NL and Reward in RL”. In *2023 Winter Simulation Conference (WSC)* <https://doi.org/10.1109/WSC60868.2023.10408669>.
- LeCun, Yann 2022. “A Path Towards Autonomous Machine Intelligence, version 0.9.2, 2022-06-27”. OpenReview. Position paper.
- SISO 2020. “SISO-STD-019-2020: Standard for Command and Control Systems – Simulation Systems Interoperation (C2SIM)”. Technical Report SISO-STD-019-2020, Simulation Interoperability Standards Organization.